KRISHISYNC AI | VIBE CODING SERIES | 5 OF 5

# Roadmap and Demo

Phases, sprints, demo script and the prompts to ship it

Market-Aware Agri-Coordination & Smart Energy Optimization for Indian Smallholders

*Challenge 1: Sustainable Agriculture & Resource Optimization*

# Roadmap, Sprints and Demo

The deck defines two phases. This plan keeps those and adds a hackathon build order and a pilot phase so the team always knows what to build next.

## Phases

| Phase | Scope | Source |
|---|---|---|
| 1 Prototyping | Core database logic, weather endpoints, anti-glut framework | Deck |
| 2 Localisation | Vernacular voice interfaces for baseline literacy access | Deck |
| 3 Pilot (proposed) | One village cluster, real sensors, real farmer feedback, measure fertiliser and peak-load changes | Proposed |

## Hackathon build order

Three workstreams run in parallel: **Architecture and backend**, **Data science and algorithms**, and **UI/UX and frontend**. The shared contract is the API shape, so agree on it in Sprint 1.

| Sprint | Goal | Architecture | Data / ML | UI / UX |
|---|---|---|---|---|
| 1 Foundation | Everything runs locally | Repo, Docker, schema, seed | Config, saturation maths, tests | Scaffold, i18n, tokens, mock API |
| 2 Core engines | Glut and irrigation work end to end | Farm API, radius query, ingestion | Anti-glut, ranking, scheduler | Onboarding, dashboard, market check |
| 3 Integrate | Real data flows to the UI | Jobs, caching, alerts | Energy, advisory, explainability | Calendar, soil, energy panel |
| 4 Hardware and demo | Judge-ready | ESP32 or simulator, deploy | Tune thresholds, demo scenario | Polish, demo mode, Hindi QA |

## Definition of done, per feature

- Runs from a clean clone using the README commands.
- Has tests for its decision logic and they pass.
- Every visible string is translated (en and hi).
- Mocked data is labelled as mocked.
- Has an empty state and an error state.

## Minimum demo-ready checklist

- Register a farm and see the Home dashboard.
- Market Check shows a RED crop and ranked alternatives with reasons.
- Rain forecast pauses a pump run; the calendar shows why.
- Energy panel shows off-peak share and an idle-load alert.
- Soil moisture changes live from a sensor or the simulator.
- Whole flow works in Hindi.

## Risks and mitigations

| Risk | Mitigation |
|---|---|
| No live mandi or regional crop data | Provider interfaces with seeded CSV; label as mock; show how real feeds plug in |
| Hardware fails on demo day | Sensor simulator script posting the same payload |
| Scope creep | Freeze after Sprint 3; only polish in Sprint 4 |
| AI-generated code drifts from spec | Pure-function engines with tests first; review every schema change |
| Weak connectivity at venue | Cached data, local Docker run as fallback |
| Claims sound unproven | Present deck figures as targets, show the explainability panel for transparency |

# Five-Minute Demo Script

| Time | Show | Say |
|---|---|---|
| 0:00 | Problem slide | Farmers plant blind. Everyone grows the same crop, prices crash, 15-20% of value is lost after harvest. |
| 0:45 | Onboarding in Hindi | A farmer registers land in under a minute, icon-first, in their language. |
| 1:30 | Market Check, RED banner | Within 20 km, this crop is over-planted. We warn before sowing, not after the glut. |
| 2:15 | Ranked alternatives and 'why' | Three market-protected options, each with a transparent reason. |
| 3:00 | Energy panel, rain pause | Rain is forecast, so the pump pauses. Heavy irrigation moves to off-peak or the solar window. |
| 3:45 | Idle-load alert and live sensor | We catch vampire load and track soil moisture in real time. |
| 4:30 | Roadmap slide | Next: vernacular voice, then a village pilot to measure real savings. |

## Likely judge questions

| Question | Honest answer |
|---|---|
| Where does regional crop data come from? | Farmers register plans in-app; with enough adoption this becomes the dataset. For the demo it is seeded. |
| Is the 25% fertiliser saving proven? | It is our target. The advisory applies nutrients only when readings are out of range; the pilot will measure it. |
| Why a web app, not just SMS? | Maps, calendars and energy panels need a richer view; voice and SMS fallbacks are planned for localisation. |
| Can it scale? | Open-source stack, PostGIS radius queries, Redis caching, stateless API. |
| What about farmers without sensors? | Core glut engine and calendar work without hardware; sensors add precision. |

# Roadmap Vibe Prompts

**VIBE PROMPT 1 | Plan the next sprint**

```
Here is my current repo state: [paste tree or describe]. We are starting Sprint [N] from
the Roadmap PDF. List the smallest ordered tasks (max 8), each with: files touched, a
test or manual check that proves it works, and an estimate in hours. Flag anything that
blocks another teammate's workstream.
```

**VIBE PROMPT 2 | Demo seed scenario**

```
Write scripts/demo_scenario.py that resets the database to a known demo state: a farmer
in Hindi, 60 neighbouring farms where one crop sits at ~0.9 saturation, a forecast with
rain tomorrow morning, a pump with an injected 9 W idle draw, and a drying soil-moisture
trend. Add a --step flag that advances the scenario so I can trigger each demo moment
on cue.
```

**VIBE PROMPT 3 | Deployment**

```
Prepare a one-command deployment: production Dockerfiles, env var handling, HTTPS via
Caddy or Nginx, and a free-tier hosting plan (for example a single small VM). Include a
smoke-test script that hits /health, registers a farm, and fetches glut-map. Write a
README section titled 'If the venue wifi dies' that explains running everything locally.
```

**VIBE PROMPT 4 | Localisation (Phase 2 starter)**

```
Extend i18n beyond en and hi: add a language-pack template and a script that finds missing
keys per language. Prototype a voice read-out for alerts using the browser Web Speech API
(speechSynthesis) in Hindi, with a graceful fallback when the voice is unavailable. Keep
alert sentences short and free of jargon.
```

**VIBE PROMPT 5 | Final QA pass**

```
Review the whole repo against the Definition of Done in the Roadmap PDF. Produce a table of
features vs: tests pass, translated, empty/error states, mock data labelled. Fix the top five
gaps, then give me a pre-demo checklist I can tick through in 10 minutes.
```